Project	**Mansourah Massarah Gold Project**
Customer:	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **CALCULATION**

CALCULATION FOR FORMWORK AND SCAFFOLD FOR 49RA MAIN ACCESS ROAD CULVERT AT 6+444 KM

Document Number:	**MD-216-49RA-CN-CV-CAL-1002**
Revision:	**D01**
Revision Date:	**09.05.2024**
Status:	**AS BUILT**

L&T Document Number:	O19045	001	X	K	00500	452	R1

REVISION HISTORY						
Rev	**Date**	**Revision Text**	**Prepared By**	**Reviewed By**	**Approved By**	**Employer**
A01	19.11.2020	First Issue	Indraneel Mukherjee	Paramartha Som	G Srinivas	
D01	09.05.2024	AS BUILT	Indraneel Mukherjee	Paramartha Som	G Srinivas	

Sensitivity: LNT Construction Internal Use

L&T Metallurgical & Material Handling
Construction Methods & Planning Centre

PROJECT :	Mansourah Massarah Gold Plant	DOCUMENT NO.		DATE
		IN167-B6-0103		17/11/2020
TITLE :	Design & Calculation detail for Slab with Cup-lock shoring for Box Culvert (6+444)	DESIGNED	CHECKED	SHEET
		I Mukherjee	P Som	4

1. INPUT FOR DESIGN CALCULATION AND DESIGN ASSUMPTIONS

INPUT — Please fill the orange coloured boxes for calculation

Depth of slab (D) =	320	mm
Density of Concrete =	2.5	MT/m^3
For Fresh concrete =	0.1	MT/m^3
Shuttering Load =	0.03	MT/m^2
Live Load =	0.15	MT/m^2
Dead load =	0.832	MT/m^2
Total Load =	1.012	MT/m^2

COMMENTS

Consideration:

DOKAPLY Birch 18mm
DOKA H-20 Timber Beam

Walling:
DOKA Multipurpose Steel Walling

Availability of Cupl-lock and MMGP:
1. Cup-lock Standard = 2m
2. Cup-lock Ledger = 0.9m and 1.3m
(As confirmed by Site Team)
Considered Cup-lock Standard = 0.5m

Refer Annexure#1

2. CHECK FOR PLYWOOD

Dokaply Birch 18mm	
Face grain parallel to span	
2440mm Long Ply	5 Spans of 488
Bending Stiffness (EI)	4.56 kNm²/m
Moment of Resistance	2.17 kNm/m
Shear Load Resistance	6.34 kN/m
Thickness (t)	18.0 mm
Weight	12.20 kg/m²

Calculation of Plywood Span

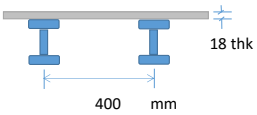
Plywood = 18 thk

PROPERTIES OF DOKAPLY Birch 18mm

Allowable Bending Moment=	0.217	MT.m/m
Allowable Shear=	0.634	MT/m
Allowable deflection=	0.0015	m
	0.002778	
Permissible EI value=	0.456	MT.m2/m
Section Modulas(Z)=	24.7	cm3/m

From bending,
Span = 1464 mm

From shear,
Span = 1253 mm



Max Spacing of Secondary Beam = 1253 mm
Max Spacing Provided = 400 mm

Check for Bending

Centre to Centre span of Plywood is 'L_1' = 400 mm

Hence, Bending Moment = $\frac{WL_1^2}{10}$

= 0.02 MT-m/m
Allowable Bending Moment = 0.217 MT-m/m

Section Modulas (Z) of 12 mm thk Plywood = 24.7 cm^3
= m^3

Bending Stress (σ_b) = $\frac{M}{Z}$

= 65.55 Kg/cm^2
Allowable Bending Stress = 879 Kg/cm^2
Hence , Safe

For Simply supported beam
Bending Moment = $\frac{WL_1^2}{10}$
Shear Force = $\frac{WL_1}{2}$
Deflection = $\frac{5WL_1^4}{384EI}$

For Continuous beam
Bending Moment = $\frac{WL_1^2}{8}$
Shear Force = $0.525(L-B-t)w$
Deflection = $\frac{0.0066WL_1^4}{EI}$

Check for Shear

Centre to Centre span of Plywood is 'L_1' = 400 mm

Hence, Shear Force = $\frac{WL_1}{2}$ MT

= 0.202 MT
Allowable Shear force = 0.634 MT
Hence, Safe

Check for Deflection

Centre to Centre span of Plywood is 'L_1' = 400 mm
Flexural Rigidity (EI) = 0.456

Max deflection = $\frac{5Wl_1^4}{384EI}$ mm

= 0.7 mm
Allowable Deflection = 1.1 mm
Hence, Safe

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3. CHECK FOR H-BEAM (Design of Secondary Beam)

COMMENTS

Calculation of H-Beam Span

Refer Annexure#2

H-Beam = H-20

UDL on H-Beam = 0.40 MT/m

From bending,

Span = 4158 mm

From Shear,

Span = 5929 mm

PROPERTIES OF H-Beam		
Allowable Bending Moment=	0.7	MT.m/m
Allowable Shear=	1.2	MT/m
Allowable deflection=	0.006	m
	0.002778	
Permissible EI value=	42	MT.m2/m

Max Spacing of Primary Beam = 4158 mm
Spacing of Primary Beam considered = 1300 mm

Check for Bending

Centre to Centre span of H beam is 'L_2' = 1300 mm

Hence, Bending Moment = $\frac{WL_2^2}{10}$

= 0.07 MT-m
Allowable Bending Moment = 0.7 MT-m
Hence, Safe

Check for Shear

Centre to Centre span of H beam is 'L_2' = 1300 mm

Hence, Shear Force = $\frac{WL_2}{2}$ MT

= 0.26 MT
Allowable Shear force = 1.20 MT
Hence, Safe

Check for Deflection

Centre to Centre span of H beam is 'L_2' = 1300 mm
Flexural Rigidity (EI) = 42 MT-m^2

Max deflection = $\frac{5WL_2^4}{384EI}$ mm

= 0.4 mm
Allowable Deflection = 3.61 mm
Hence, Safe

4. CHECK FOR STEEL WALLING (Design of Primary Beam)

Centre to Centre span of H-Beam is 'L_2' = 1300 mm
Udl on Waler = 1.32 MT/m

Refer Annexure#2

Calculation of waler Span

From bending,

Span = 2784 mm

From Shear,

Span = 15719 mm

STD. STEEL WALER		
Allowable Bending Moment=	1.02	MT.m
Allowable Shear=	10.34	MT
Allowable deflection=	0.003	m
	0.002778	
Permissible EI value=	78.4	MT.m2

Max Span of waler = 2784 mm
Span of waler = 1300 mm

Check for Bending

Centre to Centre span of Steel Walling is 'L_3' = 1300 mm

Hence, Bending Moment = $\frac{WL_3^2}{10}$

= 0.22 MT/m
Allowable Bending Moment = 1.02 MT/m
Hence, Safe

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Check for Shear

Centre to Centre span of Steel Walling is 'L_3' = 1300 mm

Hence, Shear Force = $\frac{WL_3}{2}$ MT

= 0.86 MT

Allowable Shear force = 10.34 MT

Hence, Safe

Check for Deflection

Centre to Centre span of Steel Walling is 'L_3' = 1300 mm

Flexural Rigidity (EI) = 78.4 MT-m^2

Max deflection = $\frac{5WL_3^4}{384EI}$ mm

= 0.6 mm

Allowable Deflection = 3.6 mm

Hence, Safe

COMMENTS

5. CHECK FOR CUP-LOCK LEG LOAD CAPACITY

Influence area for each leg = 1.69 m^2

Load in one Leg = 1.71 MT

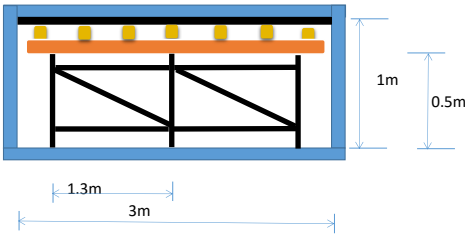
Allowable Load in leg Considered = 3.5 MT

Hence, Safe

Considereing Allowable Leg Load 6MT
Considereing Factor of Safety = 2

FINAL RESULTS FOR BOX CULVERT SLAB CASTING DESIGN WITH CUP-LOCK SCAFFOLDING:

Plywood	DOKAPLY Birch 18mm	Safe
H Beam (400mm Spacing)	DOKA H-20 Timber Beam	Safe
Walling (1300mm Spacing)	DOKA Multipurpose Steel Walling	Safe
Cup-lock Leg Load	2.63 MT	Safe
Cup-lock Bay Length	20nos x 1.3m = 26m	Safe
Cup-lock Scaffolding Width	1.3m	Safe
Number of Cup-lock Scaffolding Bay (considering 3m width per bay)	2 nos.	Safe
Cup-lock Standard Height	0.5m	Safe

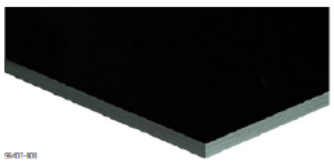


Annexure#1

The Formwork Experts.

DokaPly Birch

Data Sheet



DokaPly Birch is a birchwood-veneer plywood sheet with film coating on both sides, for horizontal and vertical formwork applications.

Sheet structure

- Film-coated plywood sheet made of Scandinavian birchwood.
- The arrangement of the veneers is crosswise.

Glue-bonding

- Boil-resistant, alkali-resistant, water-resistant and weather-resistant phenolic-resin glue-bonding.
- The glue-bonding meets the requirements of EN 314-2 Service Class 3, DIN 68705 BFU 100 or BS 6566 WBP.

Surfaces

- Both sides phenolic-resin film coating with 120 g (SC) or 220 g (DC) per m² and side.
- Edge sealing: Acrylic-resin varnish.

Technical data

Note:
All values in the tables are based on a sheet moisture content of 10 ± 2% on delivery.
Changes in the wood's moisture content can have effects on the weight, dimensions and mechanical properties of the sheet.
The grain of the outside layers of this formwork sheet runs transverse to the longitudinal direction of the sheet.

Thicknesses, weights and formats:

Nominal thickness [mm]	Layers	Weight [kg/m²]	Format [cm]
9	7	6.1	125 x 250
12	9	8.2	125 x 250 150 x 300
15	11	10.2	125 x 250
18	13	12.2	62.5 x 250 125 x 250 122 x 244 150 x 300
21	15	14.3	62.5 x 250 125 x 250 122 x 244 150 x 300

Format tolerances:

	Tolerance
Length/Width	± 3.5 mm (as per EN 315)
Perpendicularity	±1.0 mm/m
Straightness of sheet edge	±1.0 mm/m

Mechanical properties
(as per Handbook of Finnish Plywood):

Nominal thickness [mm]	E_m [N/mm²] ∥	E_m [N/mm²] ⊥	f_m [N/mm²] ∥	f_m [N/mm²] ⊥	EI [kNm²/m] ∥	EI [kNm²/m] ⊥
9	11365	6105	45.6	32.1	0.73	0.39
12	10719	6781	42.9	33.2	1.54	0.98
15	10316	7184	41.3	33.8	2.79	1.94
18	10048	7452	40.2	34.1	4.58	3.39
21	9858	7642	39.4	34.3	6.97	5.40

E_m ... mean flexural modulus of elasticity
f_m ... characteristic flexural stiffness
EI ... Flexural strength
∥ ... parallel to the grain
⊥ ... at right angles to the grain

- **Fire behaviour:** D - s2, d0
- **Thermal conductivity:** 0.17 W/mK
- **Formaldehyde class:** E1

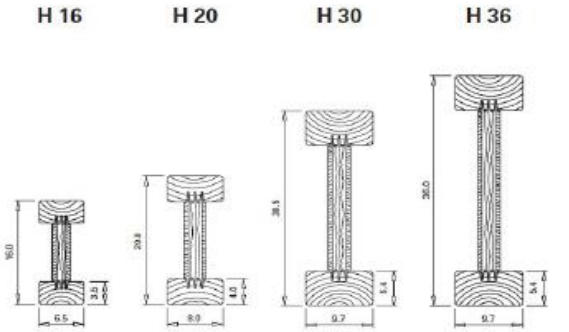
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Construction Methods & Planning Centre

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Annexure#2

Doka timber formwork beams



Dimensions in cm

Design values

	H 16	H 20	H 30	H 36	
max. permitted Q	8.5	11.0	15.0	17.0	kN
max. permitted M	2.7	5.0	13.5	17.0	kNm
E x J	250	450	1250	1850	kNm²
max. support spacing	3.20	4.00	6.00	6.00	m

Annexure#3

Load-bearing capacity of Doka standard profiles

Doka standard profiles (Multi-purpose walings WS10, WU12, WU14 and WU16) can also be used for custom constructions.

- The data given below relate to plastic boundary internal forces. This means that the specified interaction data must be complied with.
- The decreased load-bearing capacities in the end zones of the WS10 and WU12 profiles are only relevant for special applications (e.g. flexurally rigid special connecting plates), but not when used on Large-area formwork Top 50.

In order to increase the operational safety of flexurally rigid connecting devices, the following maximum load-bearing capacities shall apply:

Connecting devices	$M_{perm.}$ [kN]
Corner connecting plate SK with WS10	10.0
Corner connecting plate SK with WU12	12.0
Waling connector SL-1 WU16 0.75m with WS10	10.0
Waling connector SL-1 WU16 0.75m with WU12	15.0
Waling connector SL-1 WU16 0.75m with WU14	15.0
Waling connector SL-1 WU16 0.75m with WU16	28.0
Splice plate SKE50 plus with WS10	10.0
Splice plate SKE50 plus with WU12	11.0
Connecting plate SKE100 plus with WU14	22.0

When calculating Doka standard profiles with the Doka software 'Tipbeam', select 'Multi-purpose walings', not double U-sections.

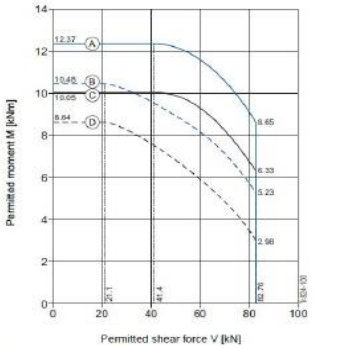
Multi-purpose waling WS10 Top50



a ... 30 cm

	Middle zone	End zone 'a'
Permitted moment M [kNm]	12.3	10.0
Permitted shear force V [kN]	82	82
Permitted normal force N [kN]¹⁾	326	268
Moment of inertia [cm⁴]	412	412

Interaction diagram



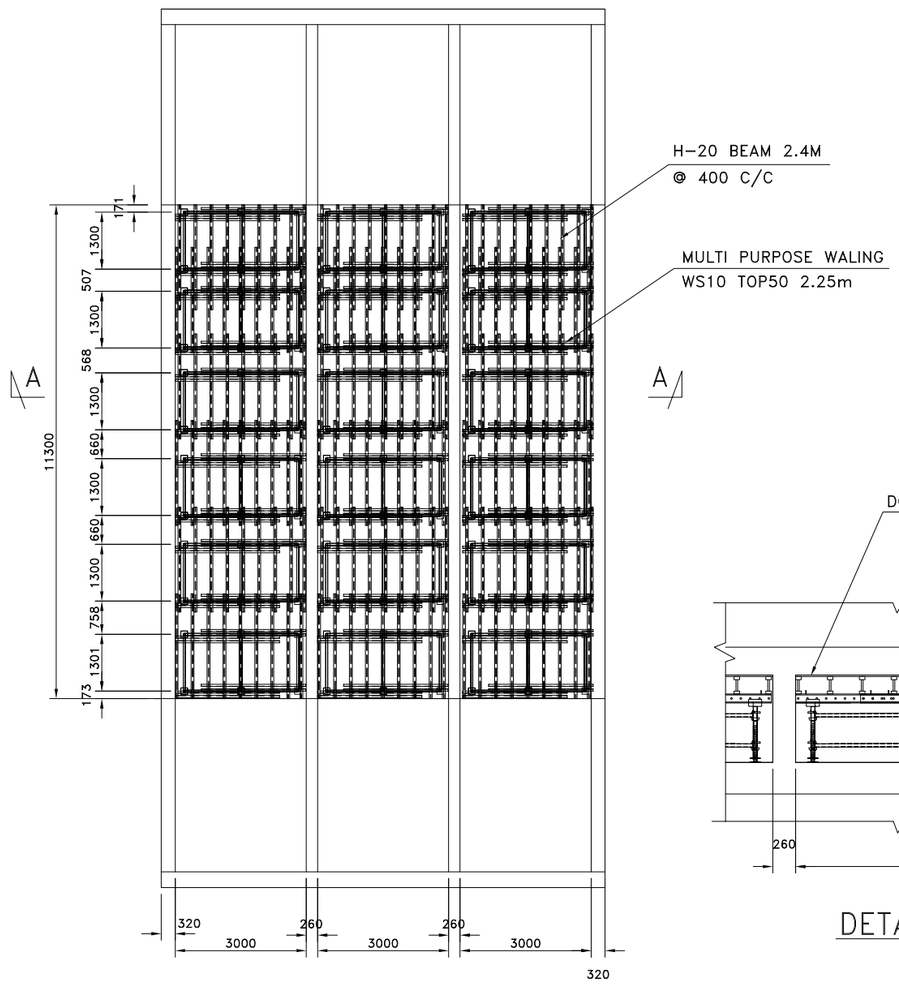
A Middle zone N_k = 0 kN¹⁾
B Middle zone N_k = 70 kN¹⁾
C End zone N_k = 0 kN¹⁾
D End zone N_k = 70 kN¹⁾

Sensitivity: LNT Construction Internal Use

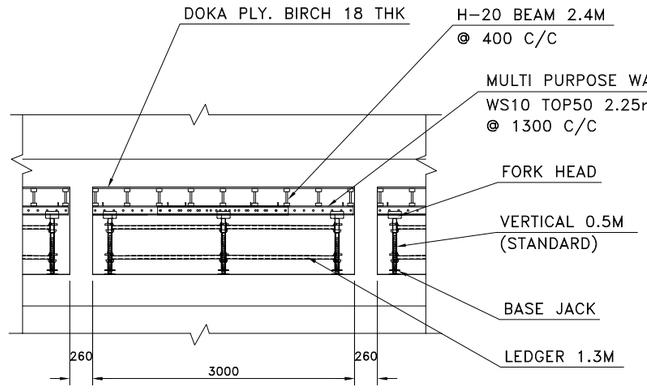
This drawing is the property of **Larsen & Toubro Limited**, ECC Division and must not be passed on to any person or body not authorised by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing

BILL OF MATERIALS

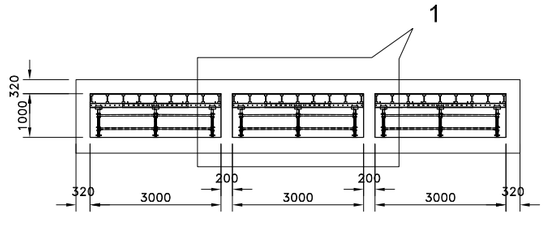
SL NO.	ITEM DISCRIPTION	QUANTITY
1	**H20 BEAM 2.4M**	**144**
2	**MULTI PURPOSE WALING WS10 TOP50 2.25M**	**72**
3	**FROK HEAD**	**108**
4	**LEDGER 1.3M**	**144**
5	**VERTICAL 0.5M (STANDARD)**	**108**
6	**BASE JACK**	**108**



PLAN



DETAIL 1



SECTION A–A

NOTE:
1. SWL for Castor wheels must not be less than 500Kg.
2. Castors should be swivel type and fixed to the standard of the tower.
3. No horizontal load should be applied to the mobile tower unless when being moved at which time the force must be applied at the base.
4. No persons should be on the platform while the mobile tower is being moved.
5. Lock all castor wheels when mobile tower is in use.
6. Always refer to TG20:08 Clause 24.12.

Fall Arrest Safety Note:
Scaffolders must remain clipped on when:
1. Moving their working platfrom e.g. when raising or lowering working platform boards,
2. Climbing up the structure to the next lift,
3. Working directly off the scaffold structure.

Load must NOT be applied to the temporary works until the structure is completed unless specifically noted otherwise.

Unloaded falsework may be vulnerable to uplift / overturning in high winds. If there is a likelihood of high winds, then the customer must provide suitable restraint to eliminate this residual hazard.

NOTE:
1. GRADE 50C Cuplok to be used.
2. Maximum Load on Working Platform is 2.0 kN/m

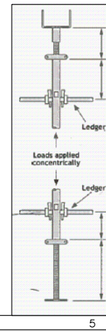
NOTE:
1. Must ensure that the Ground is adequate to safely transfer All The Loads.

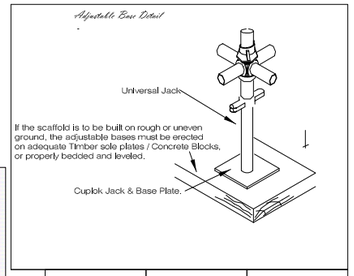
Ladder Access
1. Every sloping ladder should stand on a firm and level base,
2. Where practicable, it should be set at an angle of 4 vertical to 1 horizontal,
3. The stiles should be securely fixed to the scaffold at the top by lashing or by other suitable attachments,
4. Ladders should project at least 1.05m above the top landing place with the landing rung level with or slighlty above the level of the platform,
5. Ladders should NOT be extended by lashing two lengths together,
6. Vertical distances between two successive landings should not exceed 9.0m.

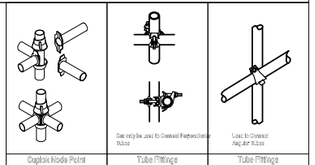
Ensure all guardposts, handrails and toeboards are fully fitted where necessary in accordance with all HSE and Site Safety Guidelines.

Bracing and lacing to be fitted as each individual section of the temporary structure is erected to avoid destabilisation. Likewise during dismantling operations the removal of the bracing MUST be carefully controlled to avoid destabilising the falsework. All bracing & lacing MUST be fully fitted prior to load being applied.

NOTE
1. **ALL DIMENSIONS ARE IN MM. & LEVELS IN M.**
2. **MINIMUM 95% OF MAXIMUM DRY DENSITY FOR BACKFILLING TO BE MAINTAINED BEFORE ERECTING CUPLOCK SYSTEM**







All Components are not shown for Clarity.

		SIGN		SIGN		SIGN	
D01	AS BUILT	DATE		DATE		DATE	
REV.NO.	DESCRIPTION	DSGN		CHKD		APPD	

REVISIONS

CLIENT: MA'ADEN Saudi Arabian Mining Company

CONSULTANT:

L&T Construction
Metallurgical & Material Handling
FORMWORK COMPETENCY CELL–KOLKATA

PROJECT: MANSOURAH MASSARAH GOLD PROJECT, KSA

TITLE: FORMWORK ARRANGEMENT FOR BOX CULVERT SLAB SHUTTER (6+444)

SCALE: 1:20 | SIZE: A1 | SHEET : 1 OF 1

	NAME	SIGN	DATE
DSGN	PRADIP HAIT		18.11.2020
CHKD	IM		18.11.2020
APPD	P.SOM		

DRG. NO.- I N 1 6 7 - B 1 - 0 1 0 5 | REV D01

RELEASED FOR ☑ PRELIMINARY ☐ TENDER ☐ APPROVAL ☐ CONSTRUCTION

\\....\A1-iso (Revised)

MA'ADEN معادن Saudi Arabian Mining Company شركة التعدين العربية السعودية	BECHTEL	**Mansourah Massarah Gold Project** **Comment Resolution Log Sheet / Compliance Report**

Outotec

LARSEN & TOUBRO

PROJECT:	**MANSOURAH MASSARAH GOLD PROJECT**	**PROJECT NO.:**	**216**		
STAGE:	**4 - PROJECT EXECUTION.**	**REVIEW LOG DOC. NO. & REV.:**			
ACONEX REFERENCE NO. & DATE:	MAADEN-STRAN-004934, 10 DECEMBER 2020				
STATUS CODE: A - APPROVED, C - APPROVED WITH COMMENTS, R - REVISE & RESUBMIT & O - OTHERS (INFORMATION, REFERENCE, REVIEWED)					

DOCUMENT NUMBER	Area / Title	Comment No.	PMC COMMENT	Comment by	EPC Response	Status	Remarks
			10-Dec-20		**05-09-24**		
MD-216-49RA-CN-CV-CAL-1002	49RA	1	Please consider comments made on WF-003461 Formwork for 49RA Main Access Road Culvert	PMC	Noted	Closed	